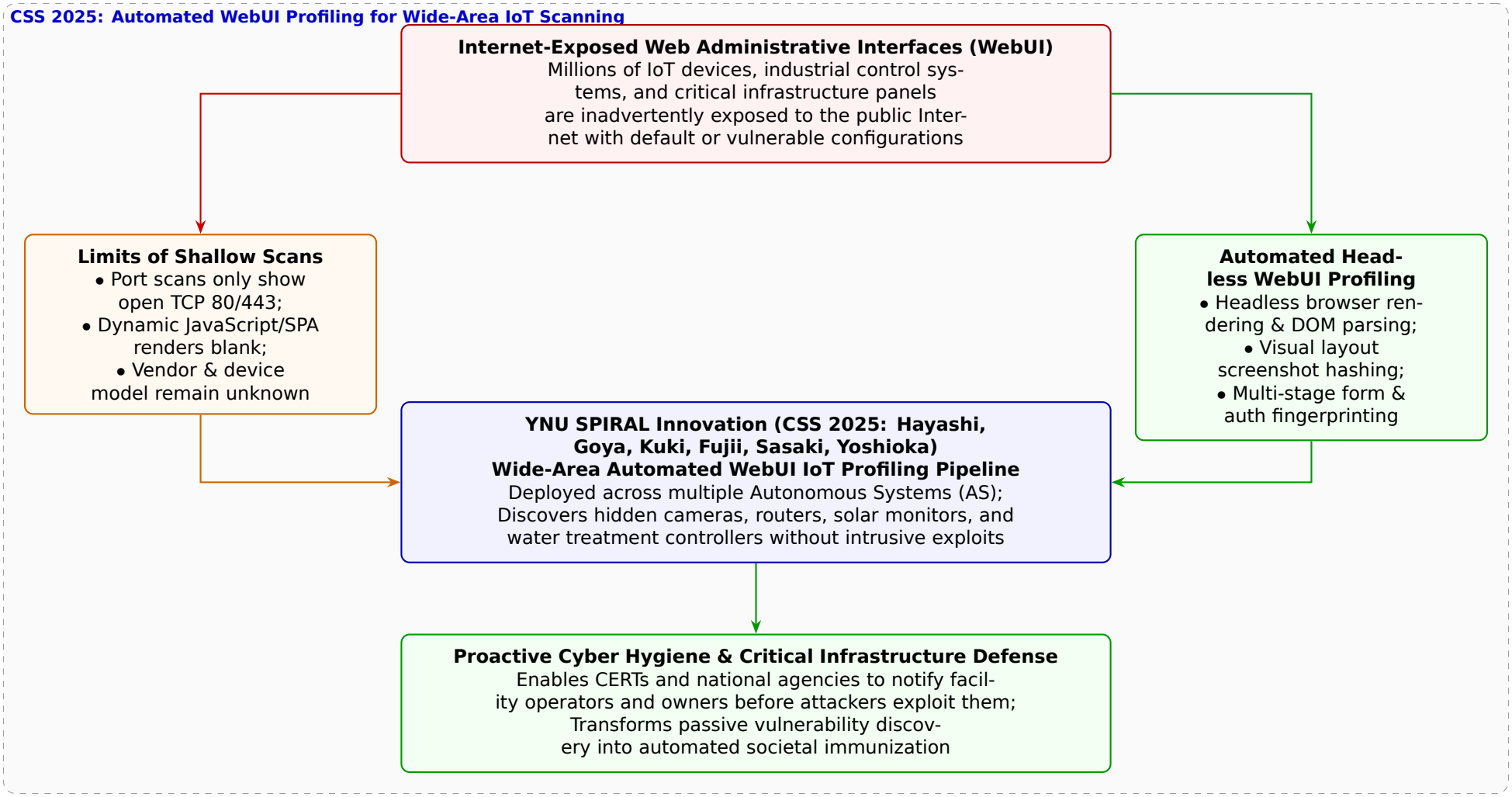
CSS 2025: Automated WebUI Profiling for Wide-Area IoT Scanning
Internet-Exposed Web Administrative Interfaces (WebUI)
Millions of IoT devices, industrial control sys-
tems, and critical infrastructure panels
are inadvertently exposed to the public Inter-
net with default or vulnerable configurations
Limits of Shallow Scans
• Port scans only show
open TCP 80/443;
• Dynamic JavaScript/SPA
renders blank;
• Vendor & device
model remain unknown
Automated Head-
less WebUI Profiling
• Headless browser ren-
dering & DOM parsing;
• Visual layout
screenshot hashing;
• Multi-stage form &
auth fingerprinting
YNU SPIRAL Innovation (CSS 2025: Hayashi,
Goya, Kuki, Fujii, Sasaki, Yoshioka)
Wide-Area Automated WebUI IoT Profiling Pipeline
Deployed across multiple Autonomous Systems (AS);
Discovers hidden cameras, routers, solar monitors, and
water treatment controllers without intrusive exploits
Proactive Cyber Hygiene & Critical Infrastructure Defense
Enables CERTs and national agencies to notify facil-
ity operators and owners before attackers exploit them;
Transforms passive vulnerability discov-
ery into automated societal immunization